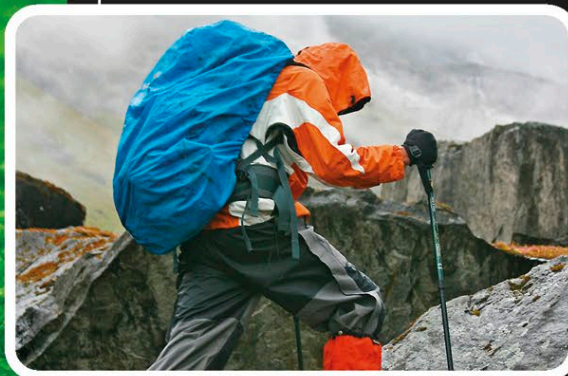


STAYING DRY

WATER-REPELLENT MATERIALS

Scientists are often inspired by nature when creating new materials, and many plants and animals have natural waterproof features that they have sought to copy. Sheep secrete an oily substance called lanolin that keeps rain off their wool, while lotus leaves have a waxy coating and a special structure that makes them especially water-resistant. By mimicking these features, scientists have created water-repellent materials that act as a barrier, so that water sits on top of them due to its natural surface tension.



HIGH AND DRY

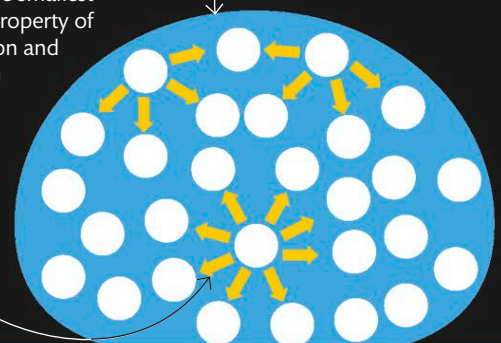
Waterproof materials are used to make clothing worn by walkers and mountaineers. One type of material, called Gore-Tex®, is breathable, as well as waterproof. It is made of many layers and has tiny holes to let out water vapor without letting in droplets of liquid water.

SURFACE TENSION

Water sits in droplets on waterproof materials, because water particles have a greater attraction to each other than they do to the air or the material's surface. As a result, water droplets shrink to form a ball, or a sphere—a shape with the smallest possible surface area. This property of water is called surface tension and can be seen in dewdrops on leaves and raindrops on windowpanes.

The forces attracting the molecules cause the water droplet to take on a spherical shape.

Water molecules are strongly attracted to each other.



The droplet sits on a waterproof surface without sinking into it.